Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Confidence Intervals

Session 14

Confidence Intervals

Session 14

Objectives:

- Construct confidence intervals (CI)
- Understand the statistical interpretation of confidence.
- Learn what factors affect the length of an interval.

What is it?

Definition: a confidence interval (CI) is a region (interval), constructed under the assumptions of our model, that contains the “true” value (the parameter of interest) with a specified probability.

A confidence interval estimates are intervals within which the parameter is expected to fall, with a certain degree of confidence

Each interval has a **confidence coefficient** (reported as a proportion): **1-α** and a **degree of confidence** (reported as a %) : **(1-α)100%**

Typical confidence coefficients are 0.90, 0.95, and 0.99, with corresponding confidence levels 90%, 95%, and 99%.

Interpretation

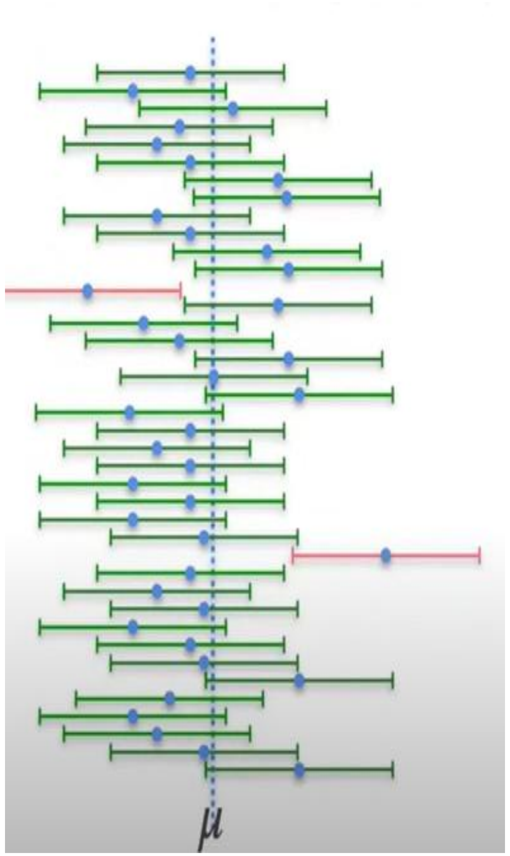
It is a common (and natural) mistake, when confronted with, for example, a 95% confidence interval, to say something like: “There is a 95% probability that the population parameter falls within confidence interval.

Why the statement is wrong:

- **The parameter is fixed** — The true mean of the population is a constant (though unknown). It either lies inside the interval or it doesn’t
- **The randomness is in the sample** — What varies is the sample we take, and therefore the interval we compute

The correct interpretation, at least for Frequentist confidence intervals, requires us to invert our thinking: instead of focusing on the probability of the parameter being in the interval, we need to focus on the probability of the interval containing the parameter.

The confidence interval contains the true population parameter approximately 95% of the time



CI estimation: Bootstrap

In order to construct a confidence interval we need information about the sampling distribution. You can construct a sampling distribution when population values were known but what happens if population values are not known?

If we have sample data, then we can use bootstrapping methods to construct **a bootstrap sampling distribution** to construct a **confidence interval**.

Definition:

Bootstrapping is a resampling procedure that uses data from one sample to generate a sampling distribution by repeatedly taking random samples from the known sample.

CI estimation: Bootstrap

Steps:

We have data concerning the heights of individuals in a random sample of n=15. To construct a bootstrap distribution for the mean height we would:

- first randomly select one individual from that sample and record their height.
- Then, with the that individual placed back into the sample, we would randomly select a second individual and record their height.

This is known as "sampling with replacement" because we are putting each case back into the sample after recording their height.

- We would repeat this process until we have selected 15 values. Because we are sampling with replacement, some individuals may appear in the bootstrap sample more than once.
- We would use those 15 selected values to compute a bootstrapped sample mean.

This process is repeated many times. The distribution of many bootstrapped sample means is known as the bootstrap distribution or bootstrap sampling distribution.

CI estimation: Bootstrap

Steps:

Finally, to obtain the CI from the boostrap distribution:

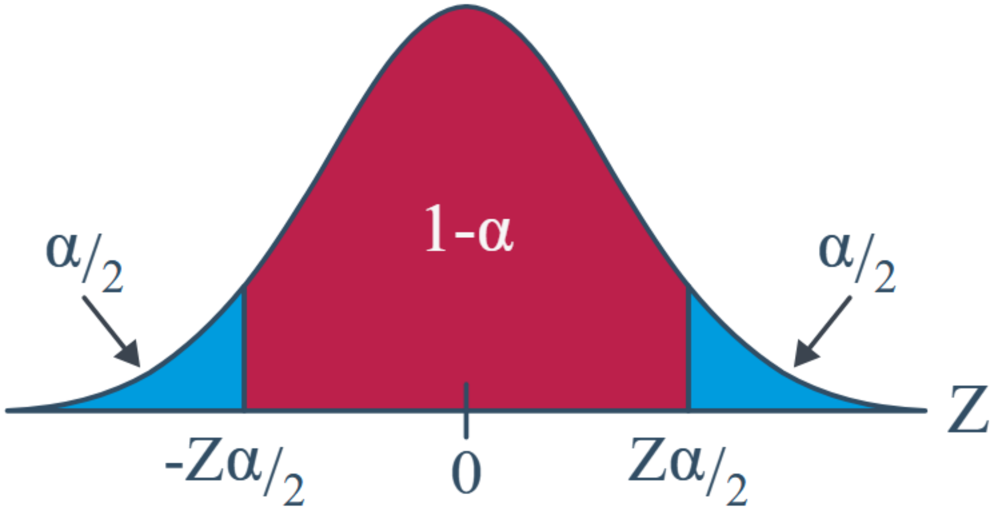
- Order all the bootstrap means from smallest to largest.
- Take the 2.5th percentile (lower bound) and the 97.5th percentile (upper bound).
- That interval is your bootstrap 95% confidence interval for the mean height.

$$\mathrm{CI}_{95\%} = \left[\hat{\mu}^*_{2.5\%},\ \hat{\mu}^*_{97.5\%}\right]$$

A Z-Interval for a Mean

Notation: $z_{\alpha/2}$ is the Z value (obtained from a standard normal table) such that the area to the right of it under the standard normal curve is $\alpha/2$.

$$P(Z \geq z_{\alpha/2}) = \alpha/2$$



A Z-Interval for a Mean

Definition

1. $X_1, X_2, \ldots, X_n$ is a random sample from a normal population with mean μ and variance σ^2. So that:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ and } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

2. The population variance $\sigma^2$is known.

Then, a $(1-\alpha)100\%$ **confidence interval for the mean** μ is:

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

The interval, because it depends on Z, is often referred to as the **Z-interval for a mean.**

A Z-Interval for a Mean

Example: A random sample of 126 persons exposed to contamination had an average blood lead level concentration of 29.2 g/dl . Assume , the blood lead level of a randomly selected person exposed to contamination, is normally distributed with a standard deviation of 7.5 g/dl.

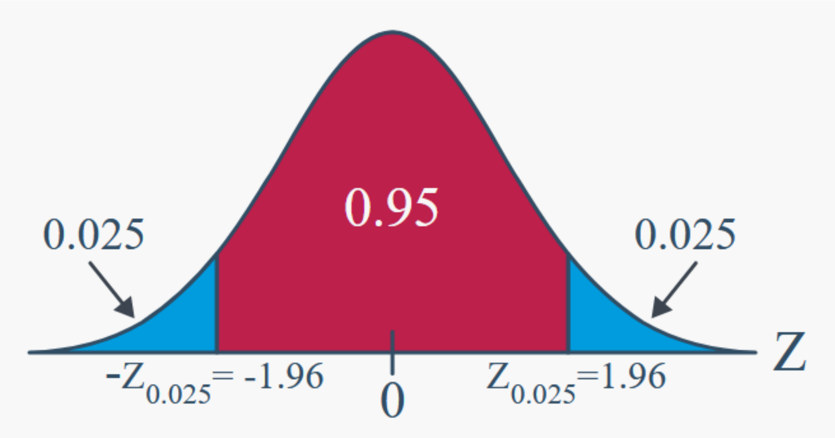
It is known that the average blood lead level concentration of humans with no exposure is 18.2 g/dl . Is there convincing evidence that the persons exposed to contamination have elevated blood lead level concentrations?

Confidence Level	Value of z_c
80% confidence	1.280
90% confidence	1.645
95% confidence	1.960
99% confidence	2.575

Let derive the 95% confidence interval, therefore 1-α=0.95 so α/2=0.025

From Z-table, $z_{\alpha/2}=1.96$

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$



$$\left[29.2 - 1.96\left(\frac{7.5}{\sqrt{126}}\right), 29.2 + 1.96\left(\frac{7.5}{\sqrt{126}}\right)\right]$$

$$[27.89, 30.51]$$

A Z-Interval for a Mean

Example: You have a population of 6000 Adults and you want to know the average weight. You select randomly 49 persons and you find a mean of 170cm and a standard deviation of 25 cm.

Calculate the 95% CI of the average height of the adults

A Z-Interval for a Mean

Solution: You have a population of 6000 Adults and you want to know the average weight. You select randomly 49 persons and you find a mean of 170cm and a standard deviation of 25 cm.

Calculate the 95% CI of the average height of the adults

Let derive the 95% confidence interval, therefore 1-α=0.95 so α/2=0.025
From Z-table, $z_{\alpha/2}$ =1.96

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

=7 cm (margin of error)

Therefore the 95% CI is: [163,177]

Imagine that you are not happy with this margin, 7cm is too much and you want only 3cm. What should be the smallest size of your random sample?

A Z-Interval for a Mean

Solution:

Imagine that you are not happy with this margin, 7cm is too much and you want only 3cm. What should be the smallest size of your random sample?

$$\bar{x} \pm z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

therefore 3=1.96*25/sqrt(n)

n>267

A t-Interval for a Mean

To use the Z-interval method we need to know the population standard deviation but what happens if YOU DO NOT KNOW IT?

We need another procedure to derive the CI.

Therefore, instead of the Z procedure we will use the T procedure and use the sample standard deviation S instead of the unknow population standard deviation.

Definition:

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom.

A t-Interval for a Mean

Definition:

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a **100(1 − α)% confidence interval on μ** is given by

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n} \qquad (8\text{-}16)$$

where $t_{\alpha/2,n-1}$ is the upper $100\alpha/2$ percentage point of the t distr $n-1$ degrees of freedom.

$$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

in mind, we say that:

1. $\bar{x}$ is a "**point estimate**" of μ
2. $\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ is an "**interval estimate**" of μ
3. $\frac{s}{\sqrt{n}}$ **is the "standard error of the mean"**
4. $t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ **is the "margin of error"**

A t-Interval for a Mean

Example: The sample mean and standard deviation of a sample of n=22 are 13.71, and 3.55 respectively.
We want to find a 95% CI on population mean

Degrees of freedom *(df)*	.2	.15	.1	.05	.025	.01	.005	.00
1	3.078	4.165	6.314	12.706	25.452	63.657	127.321	636.619
2	1.886	2.282	2.920	4.303	6.205	9.925	14.089	31.599
3	1.638	1.924	2.353	3.182	4.177	5.841	7.453	12.924
4	1.533	1.778	2.132	2.776	3.495	4.604	5.598	8.610
5	1.476	1.699	2.015	2.571	3.163	4.032	4.773	6.869
6	1.440	1.650	1.943	2.447	2.969	3.707	4.317	5.959
7	1.415	1.617	1.895	2.365	2.841	3.499	4.029	5.408
8	1.397	1.592	1.860	2.306	2.752	3.355	3.833	5.041
9	1.383	1.574	1.833	2.262	2.685	3.250	3.690	4.781
10	1.372	1.559	1.812	2.228	2.634	3.169	3.581	4.587
11	1.363	1.548	1.796	2.201	2.593	3.106	3.497	4.437
12	1.356	1.538	1.782	2.179	2.560	3.055	3.428	4.318
13	1.350	1.530	1.771	2.160	2.533	3.012	3.372	4.221
14	1.345	1.523	1.761	2.145	2.510	2.977	3.326	4.140
15	1.341	1.517	1.753	2.131	2.490	2.947	3.286	4.073
16	1.337	1.512	1.746	2.120	2.473	2.921	3.252	4.015
17	1.333	1.508	1.740	2.110	2.458	2.898	3.222	3.965
18	1.330	1.504	1.734	2.101	2.445	2.878	3.197	3.922
19	1.328	1.500	1.729	2.093	2.433	2.861	3.174	3.883
20	1.325	1.497	1.725	2.086	2.423	2.845	3.153	3.850
21	1.323	1.494	1.721	2.080	2.414	2.831	3.135	3.819

A t-Interval for a Mean

Solution: The sample mean and standard deviation of a sample of n=22 are 13.71, and 3.55 respectively.
We want to find a 95% CI on population mean

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \le \mu \le \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n}$$

n-1=21
$t_{\alpha/2,n-1}$=2,080

[12.14,15.28]

A t-Interval for a Mean

Exercice :

Determine the t-percentile that is required to construct each of the following two-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10
- Confidence level 95%, degrees of freedom 20

Determine the t-percentile that is required to construct each of the following one-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10
- Confidence level 95%, degrees of freedom 20

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

A t-Interval for a Mean

Solution : Two sided

- Confidence level 95%, degrees of freedom 10

1-α=0.95 so α/2=0.025 (two sided)

$t_{\alpha/2,n-1}$=$t_{0.025,10}$=2,228

- Confidence level 95%, degrees of freedom 20

$t_{\alpha/2,n-1}$=$t_{0.025,20}$=2,086

one-sided confidence intervals:

- Confidence level 95%, degrees of freedom 10

1-α=0.95 so α=0.05 (one sided)

$t_{\alpha,n-1}$=$t_{0.05,10}$=1.812

- Confidence level 95%, degrees of freedom 20

$t_{\alpha,n-1}$=$t_{0.05,20}$=1.725

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	$\boxed{0.05}$	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
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20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850

A t-Interval for a Mean

Exercice : A machine produces metal rods used in an automobile suspension system. A random sample of 15 rods is selected, and the diameter is measured. The resulting data (in millimeters) are as follows:

8.24 8.25 8.20 8.23 8.24 8.21 8.26 8.26 8.20 8.25 8.23 8.23 8.19 8.28 8.24

(a) Check the assumption of normality for rod diameter.

(b) Calculate a 95% two-sided confidence interval on mean rod diameter.

(c) Calculate a 95% upper confidence bound on the mean. Compare this bound with the upper bound of the two-sided confidence interval and discuss why they are different

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	$\boxed{0.05}$	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
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A t-Interval for a Mean

Solution :

8.24 8.25 8.20 8.23 8.24 8.21 8.26 8.26 8.20 8.25
8.23 8.23 8.19 8.28 8.24

(a) Check the assumption of normality for rod diameter.

Mean=8.234
Median=8.23
Mode=8.23
Std=s=0.0252

(b) Calculate a 95% two-sided confidence interval on mean rod diameter.

$$\bar{x} \pm t_{0.025,14} \frac{s}{\sqrt{n}}$$

1-α=0.95 so α=0.05 (two sided)

$t_{\alpha/2,n-1}=t_{0.025,14}=2{,}145$

[8.220, 8.248]

(c) Calculate a 95% upper confidence bound on the mean. Compare this bound with the upper bound of the two-sided confidence interval and discuss why they are different

1-α=0.95 so α=0.05 (one sided)

$$\bar{x} + t_{0.05,14} \frac{s}{\sqrt{n}}$$

$t_{\alpha,n-1}=t_{0.05,14}=1{,}761$

8.246

In the two-sided case, we split α=0.05 across two tails (α/2=0.025 each), making the critical value larger.

In the one-sided case, the entire α=0.05 is in one tail, so the critical value is smaller

A t-Interval for a Mean

Confidence Interval - t Distribution

known $\boxed{\sigma}$? $\longrightarrow$ s

Sample

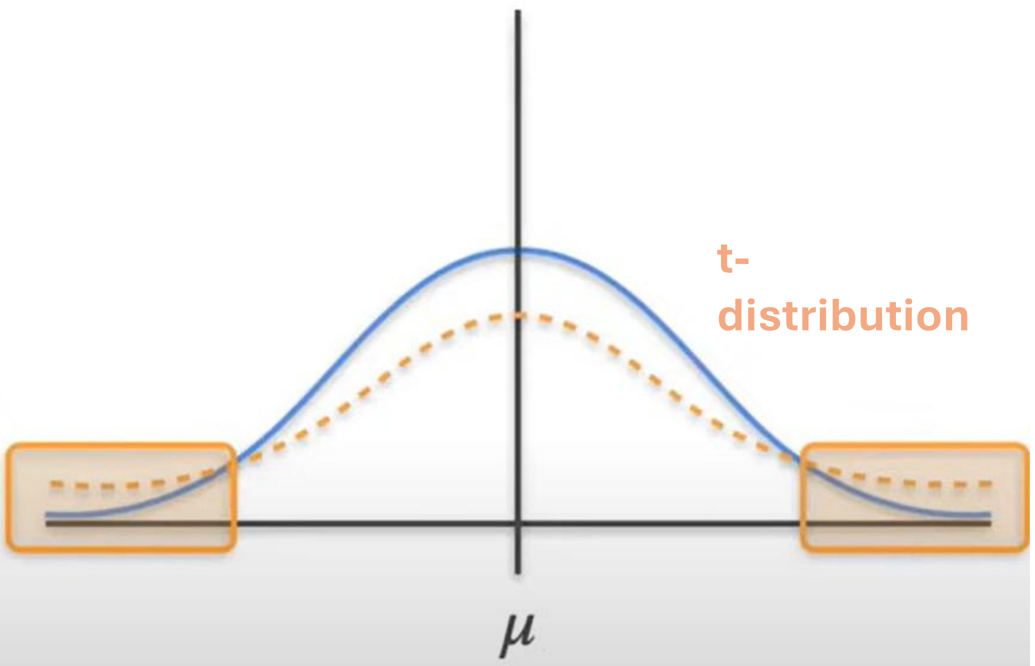
$$\bar{x} \pm \boxed{z_{1-\alpha/2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \dashrightarrow \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$$

normal distribution

not a normal distribution

student's t distribution



Normal distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable

$$X^2 = \frac{(n-1)\,S^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n - 1$ degrees of freedom.

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then **a 100(1 − α)% confidence interval on σ^2 is**

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \qquad (8\text{-}19)$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, respectively.

What to do with non normal data?

What happens if our data are skewed, and therefore clearly not normally distributed?

It is helpful to note that as the sample size n increases $T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$ approaches an approximate normal distribution regardless of the distribution of the original data

In practice!

1. Use $\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$ if the data are normally distributed.
2. If you have reason to believe that the data are not normally distributed, then make sure you have a large enough sample ($n \geq 30$ generally suffices, but recall that it depends on the skewness of the distribution.) Then:

   $$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right) \text{ and } \bar{x} \pm z_{\alpha/2}\left(\frac{s}{\sqrt{n}}\right)$$

   will give similar results.
3. If the data are not normally distributed *and* you have a small sample, use:

   $$\bar{x} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right)$$

   with extreme caution and/or use a nonparametric confidence interval for the median (which we'll learn about later in this course).

Confidence interval for two means

Definition

If $X_1, X_2, \ldots, X_n \sim N(\mu_X, \sigma^2)$ and $Y_1, Y_2, \ldots, Y_m \sim N(\mu_Y, \sigma^2)$ are independent random samples, then a $(1-\alpha)100\%$ **confidence interval** for $\mu_X - \mu_Y$, the difference in the population means is:

$$(\bar{X} - \bar{Y}) \pm (t_{\alpha/2, n+m-2}) S_p \sqrt{\frac{1}{n} + \frac{1}{m}}$$

where S_p^2, the "**pooled sample variance**":

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

is an unbiased estimator of the common variance σ^2.

Three assumptions are made in deriving the above confidence interval formula. They are:

- The measurements (X_i and Y_i) are **independent**.
- The measurements in each population are **normally distributed**.
- The measurements in each population have the **same variance** σ^2.

Confidence interval for two mean

Example:

Specie 1

sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8	sample 9	sample 10
10.2	6.9	10.9	11.0	10.1	5.3	7.5	10.3	9.2	8.8

Specie 2

sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8	sample 9	sample 10
12.9	10.2	7.4	7.0	10.5	11.9	7.1	9.9	14.4	11.3

What is the difference, if any, in the mean size of the prey (of the entire populations) of the two species?

Confidence interval for two mean

Solution:

10.2	6.9	10.9	11.0	10.1	5.3	7.5	10.3	9.2	8.8
12.9	10.2	7.4	7.0	10.5	11.9	7.1	9.9	14.4	11.3

What is the difference, if any, in the mean size of the prey (of the entire populations) of the two species?

mean_x=9.02 , std_x=1.9
mean_y=10.26, std_y=2.51
n=m=10
Sp^2=4.955 so, Sp=2.225

$$(\bar{X}-\bar{Y}) \pm (t_{\alpha/2,n+m-2})S_p\sqrt{\frac{1}{n}+\frac{1}{m}}$$

$$S_p^2 = \frac{(n-1)S_X^2+(m-1)S_Y^2}{n+m-2}$$

95% confident level.
$t_{\alpha/2,n+m-2} = t_{0.025,18}$=2,101

Therefore, the interval is: **[-0.852,3.332]**

No difference because 0 is included in the interval.

Confidence interval for two means

Definition: If the population X and Y do not have the same variance we should use the Welch's t-Interval

Welch's t-interval for $\mu_X - \mu_Y$. If:

the data are normally distributed (or if not, the underlying distributions are not too badly skewed, and n and m are large enough), and the population variances σ_X^2 and σ_Y^2 can't be assumed to be equal,
then, a $(1-\alpha)100\%$ **confidence interval for** $\mu_X - \mu_Y$, the difference in the population means is:

$$\bar{X} - \bar{Y} \pm t_{\alpha/2,r}\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$$

where the r degrees of freedom are approximated by:

$$r = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

either $\frac{s_X^2}{s_Y^2} > 4$ or $\frac{s_Y^2}{s_X^2} > 4$

Confidence interval for variances: one variance

Definition

If $X_1, X_2, \ldots, X_n$ are normally distributed and $a = \chi^2_{1-\alpha/2,n-1}$ and $b = \chi^2_{\alpha/2,n-1}$, then a $(1-\alpha)\%$ **confidence interval for the population variance** σ^2 is:

$$\left(\frac{(n-1)s^2}{b} \leq \sigma^2 \leq \frac{(n-1)s^2}{a}\right)$$

And a $(1-\alpha)\%$ **confidence interval for the population standard deviation** σ is:

$$\left(\frac{\sqrt{(n-1)}}{\sqrt{b}}s \leq \sigma \leq \frac{\sqrt{(n-1)}}{\sqrt{a}}s\right)$$

Confidence interval for variances: one variance

Exemple: A large candy manufacturer produces, packages and sells packs of candy targeted to weigh 52 grams. To estimate σ , we took a random sample of n = 10 packs off of the factory line. The random sample yielded a sample variance of 4.2 grams. Use the random sample to derive a 95% confidence interval for σ.

Degrees of freedom (*df*)	Significance level (α) .99	.975	.95	.9	.1	.05	.025	.01
1	-------	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209

Confidence interval for variances: one variance

Solution: A large candy manufacturer produces, packages and sells packs of candy targeted to weigh 52 grams. To estimate σ , we took a random sample of n = 10 packs off of the factory line. The random sample yielded a sample variance of 4.2 grams. Use the random sample to derive a 95% confidence interval for σ.

n=10 -> n-1=9 sqrt(n-1)=3
s=4.2
α =0.05
a=2.70
b=19.023

$$\left(\frac{\sqrt{(n-1)}}{\sqrt{b}} s \leq \sigma \leq \frac{\sqrt{(n-1)}}{\sqrt{a}} s \right)$$

$a = \chi^2_{1-\alpha/2,n-1}$ and $b = \chi^2_{\alpha/2,n-1}$

[1.41,3.74]

Confidence interval for variances: ratio

Definition: The confidence interval for the ratio of two variances requires the use of the probability distribution known as the ***F*-distribution.**

***F*-distribution**

If *U* and *V* are independent chi-square random variables with r_1 and r_2 degrees of freedom, respectively, then:

$$F = \frac{U/r_1}{V/r_2}$$

follows an ***F*-distribution** with r_1 numerator degrees of freedom and r_2 denominator degrees of freedom. We write $F \sim F(r_1, r_2)$.

Confidence interval for variances: ratio

Definition:

If $X_1, X_2, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$ and $Y_1, Y_2, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$ are independent random samples, and:

1. $c = F_{1-\alpha/2}(m-1, n-1) = \dfrac{1}{F_{\alpha/2}(n-1, m-1)}$ and
2. $d = F_{\alpha/2}(m-1, n-1)$,

then a $(1-\alpha)100\%$ **confidence interval** for σ_X^2/σ_Y^2 is:

$$\left(\frac{1}{F_{\alpha/2}(n-1, m-1)} \frac{s_X^2}{s_Y^2} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq F_{\alpha/2}(m-1, n-1) \frac{s_X^2}{s_Y^2}\right)$$

Confidence interval for variances: ratio

Examples:

Adult DEINOPIS	Adult MENNEUS
n = 10	m = 10
$\bar{x}$ = 10.26 mm	$\bar{y}$ = 9.02 mm
$s_X^2 = (2.51)^2$	$s_Y^2 = (1.90)^2$

Estimate, with 95% confidence, the ratio of the two population variances.

	$\alpha = 0.025$																		
	d.f.N.																		
d.f.D.	***1***	***2***	***3***	***4***	***5***	***6***	***7***	***8***	***9***	***10***	***12***	***15***	***20***	***24***	***30***	***40***	***60***	***120***	**∞**
1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.37	3.31	3.26	3.20	3.14	3.08

Confidence interval for variances: ratio

Solution:

Adult DEINOPIS	Adult MENNEUS
$n = 10$	$m = 10$
$\bar{x} = 10.26$ mm	$\bar{y} = 9.02$ mm
$s_X^2 = (2.51)^2$	$s_Y^2 = (1.90)^2$

$$\left(\frac{1}{F_{\alpha/2}(n-1, m-1)} \frac{s_X^2}{s_Y^2} \leq \frac{\sigma_X^2}{\sigma_Y^2} \leq F_{\alpha/2}(m-1, n-1) \frac{s_X^2}{s_Y^2} \right)$$

Estimate, with 95% confidence, the ratio of the two population variances.

$F_{\alpha/2(n-1,m-1)} = F_{0.025(9,9)} = 4.03$

$1/F_{\alpha/2(n-1,m-1)} = 1/4.03 = 0.248$

[0.433,7.033]

1 is included therefore, we cannot conclude that the population variances differ.

Confidence interval : one proportion

On 418 surveyed, 280 French blamed rising insurance rates on large court settlements against doctors. That is, the sample proportion is $\hat{p}=280/418$

Use this sample proportion to estimate, with 95% confidence, the parameter p , that is, the proportion of *French* who blame rising insurance rates

Confidence interval : one proportion

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Theorem

For large random samples, a $(1-\alpha)100\%$ **confidence interval** for a **population proportion** p is:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Confidence interval for variances: one proportion

Solution $\hat{p}$=280/418 .Use this sample proportion to estimate, with 95% confidence, the parameter p , that is, the proportion of *French* who blame rising insurance rates

α=0.05
$\hat{p}$=280/418=0.67
$z_{\alpha/2}$=1.96

Theorem

For large random samples, a $(1-\alpha)100\%$ **confidence interval** for a **population proportion** p is:

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

[0.625,0.715]

We can be 95% confident that between 62.5% and 71.5% of all French blame rising insurance rate

Confidence interval : two proportions

For large random samples, an (approximate) $(1-\alpha)100\%$ **confidence interval** for $p_1 - p_2$, the difference in two population proportions, is:

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Confidence interval : two proportions

Example

What is the prevalence of anemia in developing countries?

	African Women	Women from Americas
Sample size	2100	1900
Number with anemia	840	323
Sample proportion	$\frac{840}{2100} = 0.40$	$\frac{323}{1900} = 0.17$

Find a 95% confidence interval for the difference in proportions of all African women with anemia and all women from the Americas with anemia.

Confidence interval : two proportions

Solution

$\hat{p}1=0.40$
$\hat{p}2=0.17$

$\alpha=0.05$
$z_{\alpha/2}=1.96$

[20.3%,25.7%]

$$(\hat{p}_1-\hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

What is the prevalence of anemia in developing countries?

	African Women	Women from Americas
Sample size	2100	1900
Number with anemia	840	323
Sample proportion	$\frac{840}{2100}=0.40$	$\frac{323}{1900}=0.17$

Find a 95% confidence interval for the difference in proportions of all African women with anemia and all women from the Americas with anemia.

We can be 95% confident that there are between 20.3% and 25.7% more African women with anemia than women from the Americas with anemia

Confidence interval: sample size

Definition:

Estimating a population mean μ

The **sample size** necessary **for estimating a population mean** μ with $(1-\alpha)100\%$ confidence and error no larger than ϵ is:

$$n = \frac{(z^2_{\alpha/2})s^2}{\epsilon^2}$$

YOU NEED A GOOD ESTIMATE OF s^2!!!!

Exemple: A researcher wants to estimate , the mean systolic blood pressure of adult Americans, with 95% confidence and error no larger than 3 *mm Hg*. How many adult Americans (n), should the researcher randomly sample to achieve her estimation goal? From the literature, s=10

n>42

Confidence interval: sample size

Definition: The **sample size** necessary **for estimating a population proportion** of a large population with (1 α)100%) confidence and error no larger than ε:

$$n = \frac{z^2_{\alpha/2}\hat{p}(1-\hat{p})}{\epsilon^2}$$

YOU NEED A GOOD ESTIMATE OF p̂!!!!

Exemple: A pollster wants to estimate p , the true proportion of all Americans favoring the Democratic candidate with 95% confidence and error ε no larger than 0.03. From previous polls we know that p̂=80%

n=683